Hypothesis Testing (Part 1)

A statistical hypothesis test is a method of statistical inference used to decide whether the data at hand sufficiently support a particular hypothesis. Hypothesis testing allows us to make probabilistic statement about population parameters.

Null Hypothesis (H0) : is a statement that assumes there is no significant effect or relationship b/w the variables being studied. It serves as the starting point for hypothesis testing and represents the status quo or the assumptions of no effect until proven otherwise. The purpose of H0 is to gather evidence (data) to either reject or fail to reject the H0 in favour of alternate hypothesis, which claim there is a significant effect / relationship.

eg:) Lays Chips 1pkt = 100gms

H0: We challange its not 100gms, either less or more.

Alternate Hypothesis (H_1 or H_a) : is a statement that contradicts the H0 & claims there is a significant effect / relationship b/w the variables being studied. It represents the research Hypothesis or claim that the researcher want to support through stats. analysis.

eg:) H_0 : $\mu = 100gm.$

H_1 : $\mu \neq 100gms$

Important Points :

- How to decide what will be H_0 & H_1 (Typically H_0 says nothing new is happening.)
- We try to gather data/evidence to reject H0.
- It's important to note that failing to reject the H0 doesn't necessarily means that the H_0 is true; it just means that there is not enough evidence to support H_1.

Rejection region approach :

1. Formulate a H0 & H1.
2. Select a significant level (This is the probability of rejecting the H0 when it is actually true, usually set at 0.05 or 0.01)
3. Check assumption (example distribution)
4. Decide the test is appropriate (Z-test, T-test, Chi-square test, ANOVA)
5. State the relevant test statistic.
6. Conduct the test.
7. Reject or not reject H0.
8. Interpret the result.

Performing a Z-Test Eg. 1.

1) $\mu = 50$ (population mean) $\sigma = 5$ (population std)

$n = 30$ (no. of samples) $\bar{x} = 53$ (sample mean)

Step 1 : $H_0 : \mu = 50$

$H_1 : \mu > 50$

Step 2 : α (significance level) $= 0.05 \rightarrow 5\%$

Step 3 : Normality valid | σ known

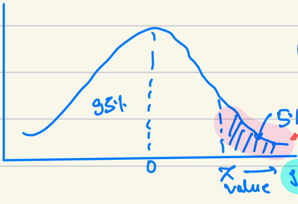
Step 4: Z-test.

Step 5: $Z = ?$

Step 6 : Z-statistic

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{53 - 50}{5 / \sqrt{30}} = 3.28$$

Step 7:



$\alpha = 5\%$

$\mu > 50$

$Z > Z\text{-value}.$

Since, $3.28 > 1.65$

So, we can reject H0

Step 8 :

$\therefore \mu > 50$

Example 2 :

$\mu = 50 \qquad n = 40 \qquad \bar{x} = 49 \qquad \sigma = 4$

$H_0 : \mu = 50$

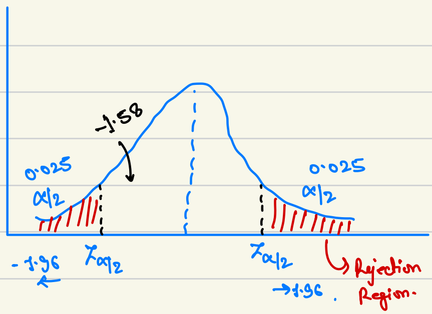
$H_1 : \mu \neq 50$

$\alpha = 5\% = 0.05$

Normality valid / σ present.

$\therefore$ Z - test

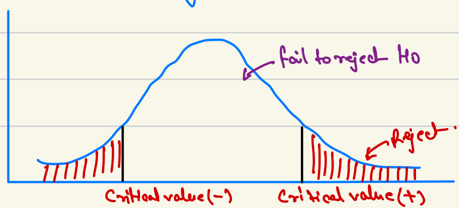
$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{49 - 40}{4/\sqrt{40}} = -1.58$$



$\therefore$ Ho can be rejected.

Significance Level : denoted as α (alpha), is a predetermined threshold used in hypothesis testing to determine whether the Ho should be rejected / not. It represents the probability of rejecting Ho when it is actually true, also known as Type 1 error.

The critical region is the region of values that corresponds to the rejection of the Ho at some choosen probability level.



Problem with rejection region approach.

- Critical value is very important even for 0.01 difference, we have to reject H0.
- Suppose z-value = 15.0 & z = 2.0, for both we have to reject H0.

  i.e. we do not have the strength measure to reject.

  ∴ We use p-value approach.

Type 1 vs Type 2 Error :

In hypothesis testing, there are 2 type of error that can occur while making H0 :-

Type I Error

Type 2 Error

Type I Error : (False Positive) error occur, when the sample results, lead to the rejection of the H0 when it is infact true.

By choosing α (significance level) a researcher can control risk of making Type I error.

Type 2 Error : (False Negative) error occur, when based on the sample results, the H0 is not rejected when it is in fact false.

This means that the researcher fails to detect a significant effect or relationship when one actually exists. The probability of comitting a Type 2 error is denoted by β.

	H0 True	H0 False
Reject H_0	Type 1 error	Correct Decision
Accept H_0	Correct Decision	Type 2 error

Power of Test $(1-\beta)$

Trade off b/w Type 1 and Type 2 error.

One sided vs. two sided test :

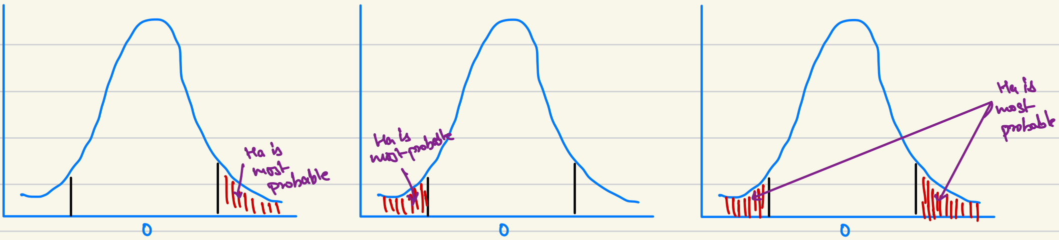
One-sided (One tailed) test : is used when a researcher is interested in testing the effect in a specific direction (either greater than or less than the value specified in H0). The H1 is on one sided test contains an inequality (either ">" or "<").

e.g.) A researcher want to test whether a new medication increases the average recovery rate compared to the existing medication.

Two-sided (two tailed) test : is used when a researcher is interested in testing the effect in both directions (i.e., whether a specified value in the H0 is different, either greater or lesser). The H1 in a two tailed test contains a "not equal to" sign. ($\neq$)

e.g.) A researcher wants to test whether a new medication has a different average recovery rate compared to the existing medication.

The main difference b/w them lies in the directionality of the H1 and how significance level (α) is distributed in the critical regions.



Ha : $\mu >$ value
Right tailed

Ha: $\mu <$ value
Left tail

Ha: $\mu \neq$ value
Two-tail test.

Two tailed test :

Adv. : Detects effects in both directions.

More conservative . : reduced Type I error.

Disadv : Less powerful : in case Ho is false

Not appropriate for directional hypothesis.

One tailed test :

Adv. : More powerful

Directional hypothesis.

Disadv : Missed effects

Increase risk of Type I error.

Uses of Hypothesis testing :

- Testing the effectiveness of interventions or treatments.
- Comparing means or proportions.
- Analysing relationships b/w variables.
- Evaluating the goodness of fit.
- A/B testing.
- Testing the independence of categorical variables.

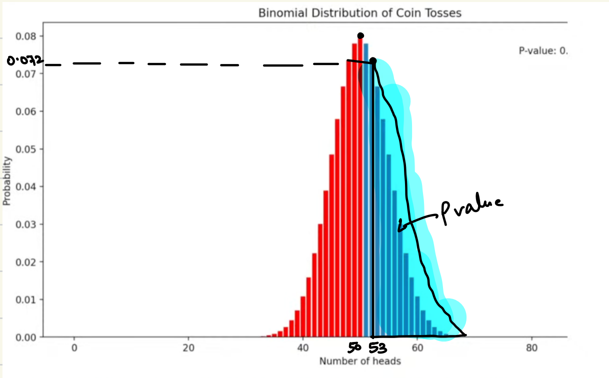
Uses in ML :

- Model Comparision
- Feature Selection.
- Hyperparameter Tuning.
- Assessing model assumptions.

Hypothesis Testing (Part 2)

P-value :

P-value is the probability of getting a sample as or more extreme (having more evidence against H0) than our own sample given the H0 is true.



← pmf

eg.) experiment 1 coin → 100 time → binomial distribution.

H0 : P(H) = P(T)

H1 : P(H) > P(T)

experiment 100 times ⟶ 53 times H.

In simple words p-value is a measure of strength of the evidence against the H0 that is provided by our sample data.

Interpreting P-value :

With α

If p-value $\leq \alpha$: Reject H0

W/o α (significance level) :

- $p < 0.01$ ⟶ Very strong evidence against H0
- $0.01 \leq p \leq 0.05$ ⟶ Moderate "
- $0.05 \leq p \leq 0.1$ ⟶ weak.
- $p \geq 0.1$ ⟶ No evidence against H0

P-value in context of Z-test :

We will use p-value approach.

Example 1 :

$\mu = 50 \quad n = 30 \quad \bar{x} = 53 \quad \sigma = 4.$

Soln.)

$H_0 : \mu = 50$

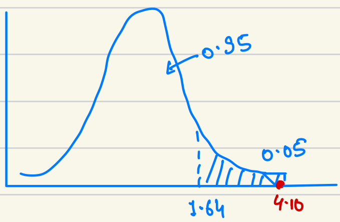
$H_1 : \mu > 50$

$\alpha = 0.05$

Normal Distr. / σ given

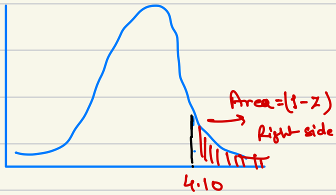
Z- statistic.

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{53 - 50}{4 / \sqrt{30}} = 4.10$$



∴ Reject H0

p-value approach :



Z-table $(4.10) \rightarrow 0.999$

Area $\rightarrow (1 - 0.999)$

p-value $\hookrightarrow 0.0001$

Since : pval $< \alpha$

Reject H0.

Example 2 :

$\mu = 50 \quad n = 40 \quad \bar{x} = 49 \quad \sigma = 5$

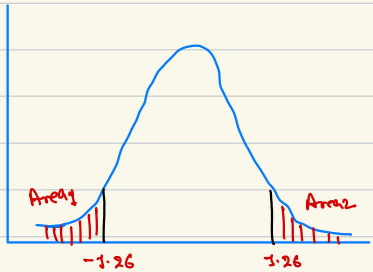
$H_0 : \mu = 50$

$H_1 : \mu \neq 50$

$\alpha = 0.05$

p-value approach :

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{49 - 50}{5 / \sqrt{40}} = -1.26$$



z-table $(-1.26) = 0.103$

p-value = Area = 2×0.103

$= 0.206$

p-value > α

$0.206 > 0.05$

Ho : Accept.

t-test :

Similar to z-test, but t-test we do not need σ, but we can use s (sample std).
Also, when we have $n < 30$.
(no. of samples).

Types of t-tests :

1. One-sample t-test : is used to compare the mean of a single sample to a known population mean. The H0 states that there is no significant difference b/w the $\bar{x}$ and μ, while H1 states there is a significant difference.

2. Independent two-sample t-test : is used to compare the means of 2 independent samples. The H0 states that there is no significant diff. b/w the means of the two samples, while the H1 states that there is significant difference.

3. Paired t-test (dependent two-sample t-test) : is used to compare the means of two samples that are dependent or paired, such as pre-test & post-test scores for the same group of subj. or measurements taken on the same subj. under two diff. conditions. The H0 states that there is no significant diff. b/w the means of the paired diff., while the H1 states that there is a significant difference.

One-sample t-test :

Assumptions :

- Normality → Normally distributed.
- Independent.
- Random sampling.
- Unknown population std. (σ)

9) $\mu = 50 \quad n = 25 \quad \bar{x} = 49.7 \quad s = 1.2$

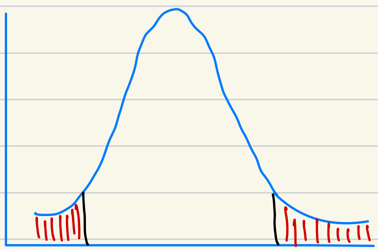
$\alpha = 0.05$

$H_0 : \mu = 50$

$H_1 : \mu \neq 50$

t-statistic.

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}} = \frac{49.7 - 50}{1.2 / \sqrt{25}} = -1.25$$



t-score $(-1.25) = 0.11$

$\therefore$ p-value $= 2 \times 0.11$

$= 0.22$

$0.22 > 0.05$

p-value $> \alpha$

H_0 : Holds.

Independent 2 sample t-test :

Assumptions :

1. Independent of observations.
2. Normality.
3. Equal variance (Homoscedasticity) $[\sigma_A^2 = \sigma_B^2] \rightarrow$ f-test | Levene test.
   if not then : Welch's test.
4. Random sampling.

eg) Desktop users : $n = 30$, $\bar{x}_A = 18.5$, $s_A = 3.5$

Mobile users : $n = 30$, $\bar{x}_B = 14.3$, $s_B = 2.7$

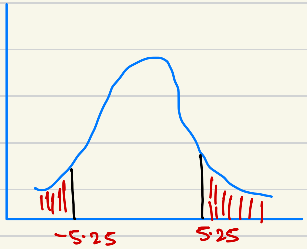
$H_0 : \mu_A = \mu_B$

$H_1 : \mu_A \neq \mu_B$

$\alpha = 0.05.$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

t-stat = 5.25.



degree of freedom = $n_1 + n_2 - 2 = 30 + 30 - 2 = 58.$ (for code)

H_0 : Reject.

Paired 2 sample test :

Common scenarios :

1. Befor & After studies.
2. Matched or correlated groups.

Assumption :

1. Paired Observation
2. Normality
3. Independent of pairs.

eg) Weight

	Wt Before	Wt. After	Diff.
A	50	55	-5
B	60	60	0
C	70	60	10
D	40	60	-20
E	25	100	-75

Normally distributed.

$\bar{x}_{diff}$

S_{diff}

$\mu_{diff} = \mu_{before} - \mu_{after}$.

$\mu_{diff} = 0$

$\alpha = 0.05$

$H_0 : \mu_{before} = \mu_{after}$

$H_1 : \mu_{before} > \mu_{after}$.

$$t = \frac{\bar{x}_{diff}}{S_{diff} / \sqrt{n}}$$

Chi-Square Tests.

Also written as χ^2 distribution, is a continuous probability distribution that is widely used in statistical hypothesis testing, particularly in context of goodness-of-fit tests & tests of independence in contingency tables. It arises when the sum of the squares of independent standard normal random variable follows this distribution.

Chi squared test has a single parameter, the degree of freedom (df), which influences the shape & spread of the distribution. The degree of freedom are typically associated with the no of independent variables or constraints in a statistical problem.

Some key property of a chi-squared distributions are:

1.) It's continuous distribution, defined for non-negative values.
2.) It is truly skewed, with the degree of skewness decreasing as the degree of freedom increases.
3.) The mean of chi-squared distribution is equal to its degree of freedom, & its variance is equal to twice the degree of freedom.
4.) As the degree of freedom increases, the chisquared distribution approaches the normal distribution in shape.

The Chi squared distribution is used in various statistical tests, such as chi squared goodness of fit test, which evaluates whether an observed frequency distribution fits an expected theoretical distribution & the Chi-square test for independence, which checks the association b/w categorical variables in a contingency table.

$\chi^2 = z^2$

normal distribution

degree of freedom (df) = 1.

for 2 $\chi^2 \Rightarrow df = 2$ $(\chi^2 = z^2 + z^2)$

3 $\chi^2 \Rightarrow df = 3$ $(\chi^2 = z^2 + z^2 + z^2)$

$$\boxed{\chi^2 = \sum_{i=1}^{k} z_k^2 \quad (df = k)}$$

df ↑ ⟶ it moves toward normal distribution.

Chi-Square Test : is a statistical hypothesis test used to determine if there is significant association b/w categorical variables or if an observed distribution of categorical data differs from an expected theoretical distribution. It is based on Chi-Square (χ^2) distribution, and it is commonly applied in two main scenarios :

Chi-square Goodness-of-fit Test : it is used to determine if the observed distribution of a single categorical variable matches an expected theoretical distribution. It's often applied to check if the data follow a specific probability distribution, such as uniform or binomial distribution.

Chi-Square Test for Independence (Chi-Square Test for Association) : It is used to determine whether there is significant association b/w two categorical variables in a sample.

χ^2 - Square Test.

Goodness of fit
(1 Categorical Col.)

Test for independence
(2 categorical diff.)

eg) Die Rolled 60 times.

	1	2	3	4	5	6
theoratical →	10	10	10	10	10	10
Observed →	5	15	5	15	5	15

↓
Goodness of fit test
↓
if observed follow uniform distri
↓
Dice is fair else unfair.

eg.) Titanic dataset → PClass (1,2,3), Survived. (1,0)

if Cols are independent or dependent.

Goodness-of-fit Test: Helps to evaluate whether the data follows any particular distribution, such as normal, possion etc.

<u>Steps</u> :—

- Define H0 and H1.
  - H0 : The observed data follows the expected theoratical distribution
  - H1 : The observed data does not follow " " ".
- Calculate the expected frequencies for each category based on the theoretical distribution & sample size.
- Compute the Chi-Square test statistic (χ^2) by comparing the observed and expected frequencies. The test statistic is calculated as:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

- O_i : Observed frequency & E_i : Expected frequency, summation is taken.
- Determine the degree of freedom (df), $df = k-1$, k = no. of categories.
- Calc. the p-value for the test statistic using the Chi-square distribution with the calculated df.
- Compare the test-statistic to the critical value or the p-value.

<u>Assumptions</u> :-

1.) Independence
2.) Categorical Data.
3.) E_i : $E_i \geq 5$ performs well.
4.) Fixed distribution : Normal, Binomial, Poisson etc should be well defined before.

*: The Chi-Square Goodness-of-fit Test is non-parametric test.

Example 1 : Die Example.

Die is fair, Die rolled 60 times.

	1	2	3	4	5	6
Oi :	12	8	11	9	10	10

Ho : Dice is fair

H1 : Dice is not fair.

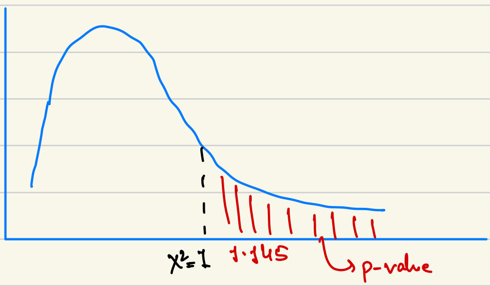
Because Dice is fair → Normal Distribution.

	1	2	3	4	5	6	Σ_i
Oi :	12	8	11	9	10	10	60
Ei :	10	10	10	10	10	10	60

$$\chi^2 = \frac{(12-10)^2 + (8-10)^2 + (10-11)^2 + (10-9)^2 + (10-10)^2 + (10-10)}{10}$$

$\chi^2 = 1$

χ^2 follow Chi-Square Distribution at $df = k-1 \rightarrow df = 6-1 = 5$



$\alpha = 0.05$

p-value = 0.96

p-value > α

Ho : accepted.

Example 2: Website owner want uniform distribution, expect equal no. of people visit / day.

	M	T	W	Th	F	S	Sun	Σ
O_i :	420	380	410	400	410	430	390	2840
E_i :	405	405	405	405	405	405	405.	

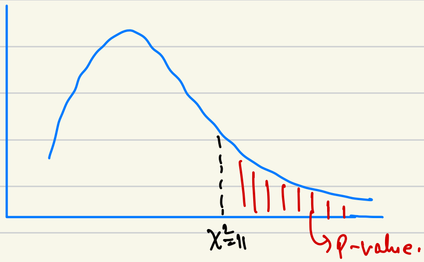
H_0: Uniform Distribution

H_1: Non uniform Distribution.

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

$\chi^2 = 11$

$df = 6$



p-value $>$ α

H_0 : Accepted.

Example 3 : A survey of 800 families in a village with 4 children each revealed the following distribution :

	Boys	: 0	1	2	3	4
O_i	Families	: 32	178	290	236	64
	E_i	:				

In this data consistent with the result that male and female births are equally probable ?

$H_0: P(m) = P(f)$

$H_1: P(m) \neq P(f)$

Bernoulli $< \begin{matrix} M \\ F \end{matrix}$ x times $\longrightarrow$ Binomial Distribution.

$p(m) = p(f) = \frac{1}{2}$

$n = 4$

$P(x) = {}^nC_x \, p^x (1-p)^{n-x}$

total # family

$P(0) = {}^4C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^3 = \frac{1}{16} \times 800 = 50$

$P(1) = {}^4C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^3 = \frac{1}{4} \times 800 = 200$

$P(2) = {}^4C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2 = \frac{\cancel{4} \times 3 \times \cancel{2!}}{\cancel{2!} \times 2!} \quad 6 \times \frac{1}{16} \times 800 = 300$

$P(3) = 200$

$P(4) = 50$

	Boys	: 0	1	2	3	4
O_i	Families	: 32	178	290	236	64
	E_i	: 50	200	300	200	50

$$\chi^2 = \frac{(32-50)^2}{50} + \frac{(178-200)^2}{200} + \frac{(290-300)^2}{300} + \frac{(236-200)^2}{200} + \frac{(64-50)^2}{50}$$

$\chi^2 =$

$df = 4$

p-value = ?

Compare p-value & α

H_0 : Accept | Reject.

Test for Independence :

Also known as Chi-Square Test for association, it's a statistical test used to determine whether there is a significant association b/w two categorical variable in sample. It helps in identifying if the occurrance of one variable is dependent on the occurrance of another variable, or if they are independent of each other.

The test is based on comparing the O_i in contingency table with E_i under the assumption of independence of each other.

Steps :-

- Formulate H0 and H1.
  - H0 : No association b/w the two categorical variables.
  - H1 : There is association b/w the two categorical variables.
- Create a contingency table with the O_i for each category of the categories of the two variables.
- Calculate E_i for each cell in the contingency table assuming that H0 is True.
- Compute the Chi-Square Statistic :

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

- $df = (n_{rows} - 1) * (n_{cols} - 1)$
- $\alpha = 0.05$
- Compare p-value & α.

Assumptions :-

- Independence of Observations.
- Categorical variables.
- Adequate sample size $(n > 5)$
- fixed marginal totals $\rightarrow \Sigma$ row and columns.

Example 1: A researcher want to investigate if there is association b/w the level of education (categorical var.) and the preferences for the particular type of exercise (cat. var.) among a group of 150 individuals. The researcher collects data and creates the following contingency table:

	Exercise Type			
Education	Yoga	Running	Swimming	Σ
School	15	20	10	45
Bachelors	20	30	15	65
Masters/Ph.D	5	15	20	40
Σ	40	65	45	150

H0: Education & Exercise are Independent.

H1: Associated.

Contingency Table — O_i

Contingency Table — E_i

	Exercise Type			
Education	Yoga	Running	Swimming	Σ
School	$\frac{40 \times 45}{150} = 12$	19	13.5	
Bachelors	17	28	40	
Masters/Ph.D	10	17	12	
Σ				

P(H) & P(6)

$\Rightarrow \frac{1}{2} \times \frac{1}{6}$

$\Rightarrow \frac{1}{12}$ (independent event.)

P(School) · P(Yoga)

$= \frac{45}{150} \times \frac{40}{150}$

For. No.

$= \frac{45 \times 40}{(150)^2} \times 150$

$= 12$

$$\chi^2 = \frac{(15-12)^2}{15} + \frac{(20-19)^2}{20} + \cdots + \frac{(20-12)^2}{12}$$

$\chi^2 =$

$df = (3-1) * (3-1) = 4$

$\overset{\rightarrow 0.04}{\text{p-value}}$ & α

H_0 : $0.04 < \alpha \rightarrow$ Reject.

Uses in Machine Learning :

1. Feature Selection.
2. Evaluation of Classification models.
3. Analysing relation b/w categorical features.
4. Discretization of continuous vars.
5. Variable selection in decision tree.

ANOVA

ANOVA → Analysis of Variance.

f- distribution :

- It's continuous probability distribution.
- Also known as Fisher-Snedecor distribution.
- Defined by 2 parameters — df1 : degree for the numerator.
  df2 : degree for the denominator.
- Positively skewed and bounded (left bound = 0)
- Test hypotheses about the equality of 2 variances in different samples & populations.
- Comparing statistical model.

$$\chi_1^2 \longrightarrow df_1$$
$$\chi_2^2 \longrightarrow df_2$$

$$\boxed{\text{f-distribution} = \frac{\chi_1^2 / df_1}{\chi_2^2 / df_2}}$$

One-way ANOVA test : is a statistical method used to compare the means of 3 or more independent groups to determine if there are any significant difference b/w them. Its extension of the t-test, which is used for comparing the means of two independent groups. The term one-way refers to the fact that there is only one independent variable (factor) with multiple levels (groups) in this analysis.

The primary purpose of one-way ANOVA is to test the H0 that all the group means are equal. The H1 is that at least one group mean is significantly different from others.

Steps :—

- Define H_0 & H_1.
- Calc. overall mean of all the groups combined and mean of all the group individually.
- Calc. the "between-group" and "within-group" sum of squares (SS).
- Find between-group & within-group df (degree of freedom)
- " " " mean squares (MS)
- " " " F-statistics.

Source of Variation	Sum of Squares (*SS*)	Degrees of Freedom (d.f.)	Mean Square (*MS*) (This is *SS* Divided by d.f.) and is an Estimation of Variance to be Used in *F*-ratio	*F*-ratio
Between samples or categories	$n_1(\overline{X}_1 - \overline{\overline{X}})^2 + \cdots + n_k(\overline{X}_k - \overline{\overline{X}})^2$	$(k-1)$	$\frac{SS \text{ between}}{(k-1)}$	$\frac{MS \text{ between}}{MS \text{ within}}$
Within samples or categories	$\sum(X_{1i} - \overline{X}_1)^2 + \cdots + \sum(X_{ki} - \overline{X}_k)^2$ $i = 1, 2, 3, \ldots$	$(n-k)$	$\frac{SS \text{ within}}{(n-k)}$	
Total	$\sum(X_{ij} - \overline{\overline{X}})^2$ $i, j = 1, 2, 3, \ldots$	$(n-1)$		

Example 1 :

A	B	C
3	1	8
6	8	6
3	9	10

Step 1 :

$H_0 : \mu_A = \mu_B = \mu_C$

H_1 : Atleast 1 of them is different.

Step 2 :

no. of samples $(n) = 9$

no. of category $(k) = 3$

$$\text{Grand mean } (\overline{\overline{X}}) = \frac{\overline{X}_A + \overline{X}_B + \overline{X}_C}{3} = 6$$

$\bar{x}_A = 12/3 = 4$
$\bar{x}_B = 18/3 = 6$
$\bar{x}_c = 24/3 = 8$

Step 3: Sum of Squares Total. (SST)

$SST = \Sigma\ \bar{\bar{x}} - n_{ij}$

$SST = (6-3)^2 + (6-6)^2 + (6-3)^2 + (6-1)^2 + (6-8)^2 + (6-9)^2$
$+ (6-8)^2 + (6-6)^2 + (6-10)^2$

$SST = 76.$

Step 4: SSW → Sum of Squares Within-group.

$\bar{x}_A = 4 \qquad \bar{x}_B = 6 \qquad \bar{x}_c = 8$

$SSW = (4-3)^2 + (4-6)^2 + (4-3)^2 +$
$(6-1)^2 + (6-8)^2 + (6-9)^2 +$
$(8-8)^2 + (8-6)^2 + (8-10)^2$

$SSW = 52.$

Step 5: SSW $\xrightarrow[\text{SSW}]{\text{with respect to}}$ df (degree of freedom)

df_{SSW} : $(n-k) \Rightarrow (9-3) = 6$

df_{SST} : $(n-1) \Rightarrow (9-1) = 8.$

Step 6: Sum of Squares Between (SSB).

$SSB = \Sigma n_i (\bar{\bar{x}} - \bar{x}_i)^2$
$= n_A(\bar{\bar{x}} - \bar{x}_A)^2 + (\bar{\bar{x}} + \bar{x}_B)^2 n_B + (\bar{\bar{x}} + x_c)^2 n_c$
$= 3 \cdot (6-4)^2 + 3 \cdot (6-6)^2 + 3\ (6-8)^2$

$SSB = 24$

Step 7 :

$$df_{SSB} = (k-1) = (3-1) = 2$$

Quantity	Value	df
SSB	24	2
SSW	52	6
SST	76	8

SST = SSW + SSB. → Always true

Step 8 : Mean Square (MS)

$$MS_B = \frac{SS_B}{df_B} = \frac{24}{2} = 12$$

$$MS_W = \frac{SS_W}{df_W} = \frac{52}{6}$$

Step 9 : f-statistic $= \frac{MS_B}{MS_W} = \frac{12 \times 6}{52} = 1.4$

Step 10 : Calc. p-value

```
import scipy.stats as stats.
f_statistic = 1.4
df1 = 2
df2 = 6
p-value = stats.f.sf (f_statistic, df1, df2)
```

p-value = 0.31

p-value > α

$\therefore$ H0 : Accept. i.e. $\mu_A = \mu_B = \mu_C$

Geometric Intuition :

$$(\bar{\bar{x}} - \bar{x}_A)^2 + (\bar{\bar{x}} - \bar{x}_B)^2 + (\bar{\bar{x}} - \bar{x}_C)^2$$

$\uparrow$ fixed $\quad$ $\uparrow$ vary random.

$\bar{\bar{x}} - \bar{x}_A \xrightarrow{\text{follow}}$ Normal distribution

$\Sigma(\bar{\bar{x}} - \bar{x}_A)^2 \xrightarrow{\text{follow}}$ Chi-Square dis.

$$F = \frac{\frac{SSB}{df_{SSB}}}{\frac{SSW}{df_{SSW}}}$$

- F will be +ve due to square.

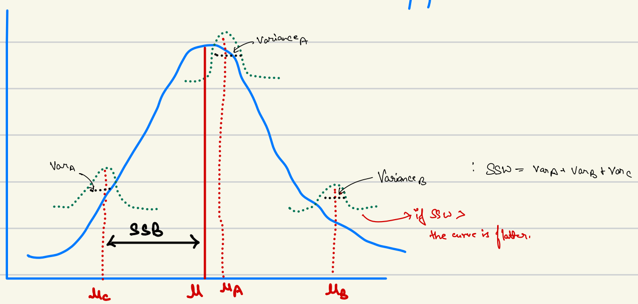
$F \uparrow \longrightarrow SSB \uparrow$ & $SSW \downarrow$.

H0: All the 3 groups come from same population.

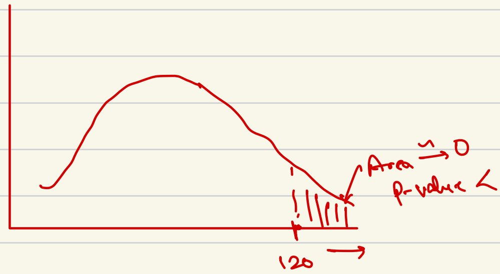
$\mu_A = \mu_B = \mu_C$

$\downarrow$

thus pop μ is same.



As SSB ↑ , maybe the groups are far away, may not be part of the population.



∴ H0 fails

Assumptions :

1. Independence
2. Normality
3. Homogenity of variances. ⟶ Otherwise ⟶ Welch's ANOVA.

Post-hoc Test :

Which value led to failure of ANOVA.
This test is performed after the ANOVA to determine which specific groups or pairs of groups have significantly diff. means.

Main purpose is to control FWER ⟶ family-wise Error Rate and adjust α for multiple comparisions to avoid Type I Errors.

Post hoc test

- Bonferroni Correction
- Turkey's HSD (High Significance Difference) test.

Bonferroni Correction :

iteration

1 AB ⟶ t-test ⟶ p-val.

2 BC ⟶ t-test ⟶ p-val.

3 CA ⟶ t-tes ⟶ p-val.

Culprit combination will have p-val. less.

Problem with test ⟶ $\alpha = 0.05$ ⟶ Type 1 error. chance increases.

$\therefore$ Divide $\alpha/3$ due to 3 iteration

↳ this is the correction.

Interview Q.

** Why t-test is not used for more than 3 categories?

1. Increased type 1 error.
2. Difficulty in interpreting results.
3. Inefficiency.

ML Applications :

1. Hyperparameter Tuning.
2. Feature Selection.
3. Algo. Comparision.
4. Model Stability assesment.